

# Linear Algebra

mat

|   | 0  | 1  | 2  | column |
|---|----|----|----|--------|
| 0 | 2  | 3  | 5  |        |
| 1 | 7  | 14 | 17 |        |
| 2 | 21 | 27 | 42 |        |

rows

↳  $\text{mat}[\text{row-index}][\text{col-index}]$

$$\text{mat}[2][1] = 27$$

Scalar  $\longrightarrow$  0-dimensional data

Vector  $\longrightarrow$  1-dimensional data

Matrix  $\longrightarrow$  2-dimensional data

## Transpose

|   | 0   | 1  | 2  |
|---|-----|----|----|
| 0 | 2   | 4  | -1 |
| 1 | -10 | 5  | 11 |
| 2 | 18  | -7 | 6  |

$\longrightarrow$

| 0  | 1   | 2  |
|----|-----|----|
| 2  | -10 | 18 |
| 4  | 5   | -7 |
| -1 | 11  | 6  |

Multiplication  $\longrightarrow$  col of 1st matrix = row of 2nd matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3} * \begin{bmatrix} 7 & 4 & 8 \\ 2 & 7 & 9 \end{bmatrix}_{2 \times 3}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3} * \begin{bmatrix} 7 \\ 2 \\ 1 \end{bmatrix}_{3 \times 1} \longrightarrow \begin{bmatrix} 1*7 + 2*2 + 3*1 \\ 4*7 + 5*2 + 6*1 \\ 7*7 + 8*2 + 9*1 \end{bmatrix}_{3 \times 1}$$

$$\underbrace{a \times b}_{a \times c} \underbrace{b \times c}$$

$$\begin{bmatrix} 14 \\ 44 \\ 74 \end{bmatrix}$$

$y = a * b \rightarrow$  element wise multiplication  
 $y = a @ b \rightarrow$  matrix multiplication

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$

$$\begin{matrix} 1 \times 1 & 2 \times 5 & 3 \times 7 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} * \begin{bmatrix} 1 & 5 & 7 \\ 2 & 4 & 6 \\ 7 & 9 & 11 \end{bmatrix} \Rightarrow \begin{bmatrix} 26 & 40 & 52 \\ 56 & 94 & 124 \\ 86 & 148 & 196 \end{bmatrix}$$

Dot Product

$$\begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 2 \\ 8 \end{bmatrix} = 2 \cdot 8 + 7 \cdot 2 + 1 \cdot 8 = 38 \quad \leftarrow \sum u_i * v_i$$

$\swarrow v_1 \quad \swarrow v_2$

Norm  $\rightarrow$  Length of a vector

$$0.7, -0.2, 0.1 \rightarrow \sqrt{(0.7)^2 + (-0.2)^2 + (0.1)^2}$$

Vectors vs tensor  $\rightarrow$  generalization of  
scalar, vector

$\rightarrow$  1D data

$\hookrightarrow$  magnitude (Norm)

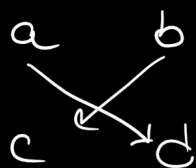
$\hookrightarrow$  Direction

& matrix

|               | type of data | tensor order |
|---------------|--------------|--------------|
| scalar        | 5            | 0            |
| vector        | [1, 2, 3]    | 1            |
| matrix        | 2x2          | 2            |
| Higher tensor | 3D array     | 3 +          |

$\rightarrow$  cosine  $\rightarrow \frac{A \cdot B}{\|A\| \|B\|}$

$\rightarrow$  way to check how similar the  
two vectors are



$ad - bc$

1 2  
3 4

$\rightarrow 4 - 6 = -2$

$$A \rightarrow \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

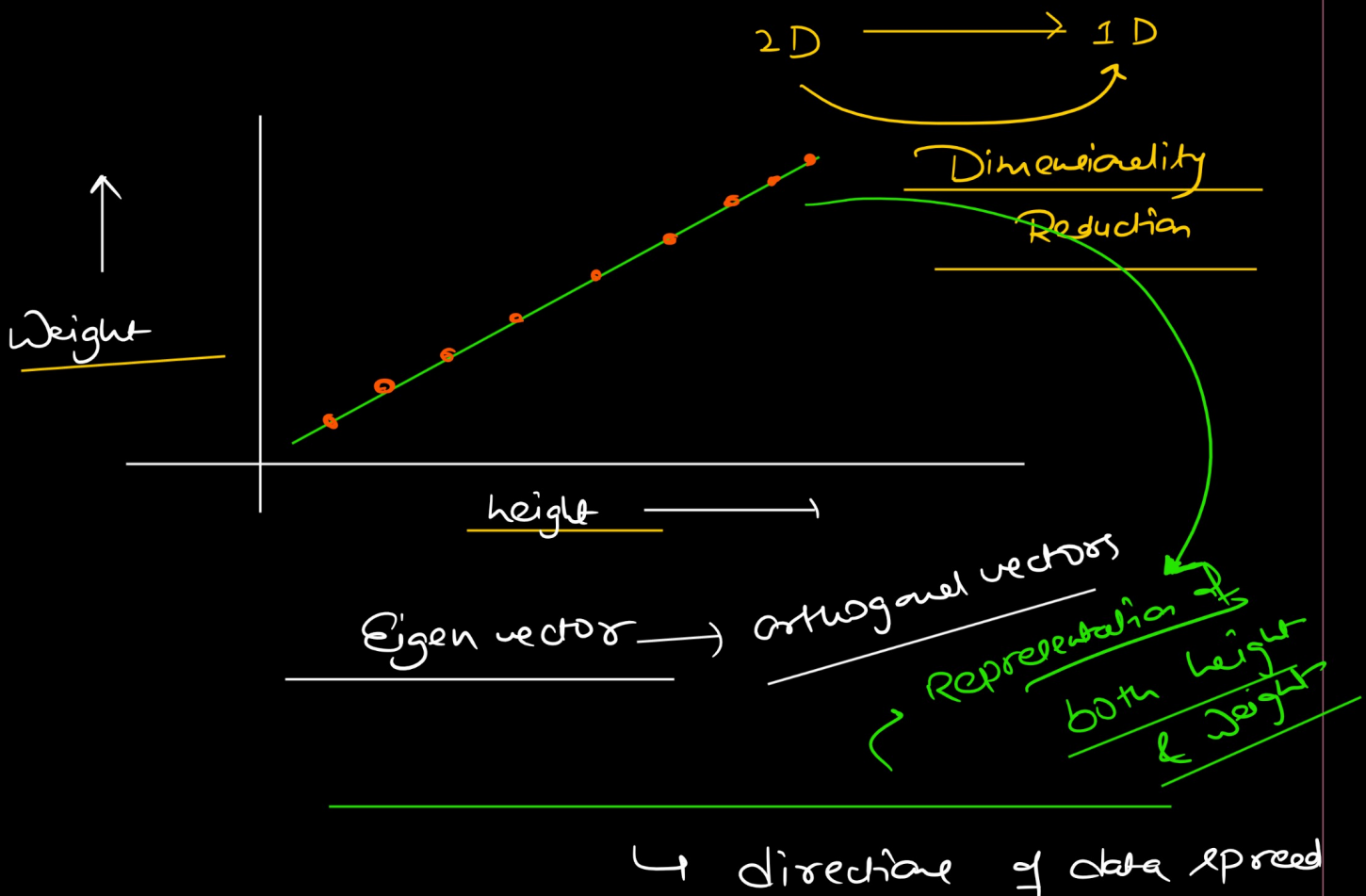
$$A^T \rightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0*0+1*1 & 0*-1+0*0 \\ -1*0+0*1 & -1*-1+0*0 \end{pmatrix}$$

$$\leftarrow A \cdot A^T = I \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A^{-1} = A^T$$

$$\frac{1}{A} = A^T$$



Eigen value  $\rightarrow$  importance of those

$A X = \lambda X$  dis chap

$\swarrow$  eigen vector

$\searrow$  eigen value

$\swarrow$  original

matrix

$$A \rightarrow \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 2-\lambda & 0 \\ 0 & 3-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)(3-\lambda) = 0$$

$$6 - 2\lambda - 3\lambda + \lambda^2 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\lambda^2 - 3\lambda - 2\lambda + 6 = 0$$

$$\lambda(\lambda - 3) - 2(\lambda - 3) = 0$$

$$\lambda = 2, 3$$

eigen vector

$$\lambda = 2$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = 1, y = 0$$

$$\lambda = 2 \longrightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

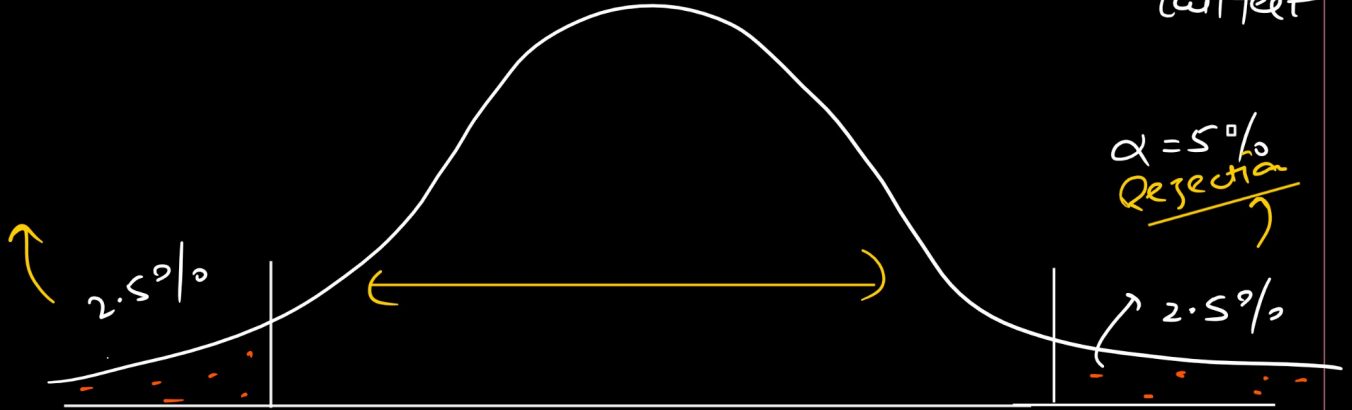
$$x = 0, y = 1$$

$$\lambda = 3 \Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$H_0 \rightarrow \mu_{\text{sample}} = \mu_{\text{pop}}$$

$$H_1 \rightarrow \mu_{\text{sample}} \neq \mu_{\text{pop}}$$

two  
tail test



$$H_0 \rightarrow \mu_{\text{sample}} = \mu_{\text{pop}}$$

$$H_1 \rightarrow \mu_{\text{sample}} > \mu_{\text{pop}} \rightarrow \text{Right tailed}$$

